"Matrix Algebra"

যোগ, বিয়োগ, গুণ, ভাগ নিজে করতে নিব।

Transpose Matrix:

Row হবে Column

Column হবে Row

Conjugate Matrix:

$i+5x \xrightarrow{\text{হবে}} -i+5x$

$$A = \begin{bmatrix} 1 & i & 1-i \\ -i & 2 & 2+3i \\ 5 & 1+2i & -5 \end{bmatrix}$$

$$\bar{A} = \begin{bmatrix} 1 & -i & 1+i \\ i & 2 & 2-3i \\ 5 & 1-2i & -5 \end{bmatrix}$$

Conjugate Transpose (Hermetian) Matrix:

$$A^* = (\bar{A})^T = A$$

Orthogonal Matrix:

$$AA^T = A^T A = I$$

Idempotent Matrix:

$$A^2 = A$$

Involutory Matrix:

$$A^2 = I$$

Inverse Matrix:

$$A^{-1} = \frac{1}{|A|} \text{Adj } A$$